

**INTEGRATED CLIMATE-INTELLIGENT SALT
PRODUCTION ECOSYSTEM: A MULTI-MODAL AI
FRAMEWORK FOR SUSTAINABLE YIELD
OPTIMIZATION IN TROPICAL COASTAL
ENVIRONMENTS**

Project ID: 25-26J-431

Project Proposal Report

Sirimanna RDIB – IT22308016

Supervised by: Dr. Nathali Silva

Co-supervised by: Mrs. Thilini Jayalath

B.Sc. (Hons) Degree in Information Technology, specializing in
software engineering

Department of Software Engineering
Faculty of Computing

Sri Lanka Institute of Information Technology

Sri Lanka

August 2025

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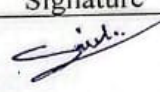
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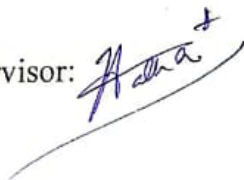
DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Date: 2025/08/22

Signature of the Co-Supervisor:



Date: 22/08/2025

ABSTRACT

Crystallization Forecasting module of the Integrated Climate-Intelligent Salt Production Ecosystem is relevant in overcoming a serious problem in the traditional salt-making industry in Sri Lanka, namely, disrupted salt-making as uncertainty mounts in the weather pattern. Solar salt production process, which depends on constant weather conditions, is susceptible to shocks by unseasonal rainfall, changes in temperatures, and storms, that unsettle the production and negatively affect the yield reliability, and economic sustainability. The study set out to provide accurate, localized, and timely forecast of the best time to crystallize the salt based on a combination of historical production data, receipt of weather records, and satellite remote sensing data.

We approach this process in a very systematic manner, including a multi-staged pipeline consisting of the processes of data collection, preprocessing, advanced machine learning, deep learning model development, and visualization. Data is collected via digitized handwritten logbooks; national and global climate data sets, as well as satellite imagery data capturing environmental and operating parameters that affect crystallization. The information is then cleaned thoroughly, normalized, and engineered into effective indices like evaporation potential and salinity conditions to help train solid models

The forecasting engine consists of a multi-model architecture based on classical statistical time series (ARIMA, Prophet) and machine learning regressors (Random Forest, XGBoost) and deep neural sequence models (LSTM, GRU). Ensemble techniques are also used to increase predictive accuracy and robustness. Model performance is measured based on back-testing against huge historical datasets by use of indicators such as RMSE and MAE, where hyper-parameter tuning is used to enhance predictive power.

The forecast results are presented in a user-friendly interface with daily to seasonal readiness forecasts, an easy-to-interpret traffic-light alert, highly visual trend graphs, and risk alerts. This enables operators to make harvesting calls based on data which significantly reduces losses due to weather and increases economic benefit.

The commercialization potential of this component is the impact it has on changing the traditionally reactive salt production process into a proactive, AI-enabled one that is responsive to microclimates in each area. Based on advanced analytics, high-quality user interfaces, and extensive domain expertise, the Crystallization Forecasting component is an innovative, effective tool that will make climate-sensitive salt industries more resilient, efficient, and sustainable. Such innovation contributes to the self-sufficiency agenda of Sri Lanka, the coastal economy, and provides a model that can be adopted to produce salt worldwide in the light of climate change demands.

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LIST OF ABBREVIATIONS

Table 0.1: List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
ARIMA	Autoregressive Integrated Moving Average
CSV	Comma-Separated Values
DaaS	Data-as-a-Service
DL	Deep Learning
EMA	Exponential Moving Average
ESA	European Space Agency
GPU	Graphics Processing Unit
GRU	Gated Recurrent Unit
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
ML	Machine Learning
MODIS	Moderate Resolution Imaging Spectroradiometer
NOAA	National Oceanic and Atmospheric Administration
PDF	Portable Document Format
RMSE	Root Mean Square Error
SaaS	Software-as-a-Service
SQL	Structured Query Language
SSA	Singular Spectrum Analysis
UI	User Interface

1. INTRODUCTION

The Puttalam salterns on the northwestern coast of Sri Lanka are the basis of the solar salt industry in the country, which is powered by large shallow ponds and conventional solar evaporation. These traditional methods, although cheap and eco-friendly when conducted in a country with a stable climate, become highly susceptible to the worsening conditions of climate change. The last few years have produced highly sporadic rain patterns, high and low temperatures, along with unseasonal storms, which have disrupted the evaporation patterns, causing a fall in the predictability of yield, and the self-sufficiency of salt may be in trouble with the nation. The effects of such disruptions travel across the local economy-employment, export potential, and food security. [1]

Artificial intelligence (AI) and machine learning (ML) have changed industrial planning and agriculture around the world. Use cases involve predictive climate analytics, computer-vision-based automated quality control, and distributed optimization through federated learning. Interestingly, recently, there has been the introduction of AI-enabled platforms, which involve the use of weather information, sensors, and even satellite measurements to offer decisive information and dynamic guidance in decision-making both to the salt producers and also to the crop producers. But the effective use of the said technologies in the salt industry, particularly the Puttalam salterns in Sri Lanka, is practically non-existent, thus bringing to light a specific research and innovation gap.

In Puttalam, most of the Sri Lankan salterns still rely on fragmented, empirical management procedures, even though solutions such as Salt Bae demonstrate the efficacy of AI-powered prediction algorithms that smallholder salt-farmers can deploy daily, and which leverage multimodal data to drive highly accurate and timely production recommendations. This gap only highlights the need to have a comprehensive decision-support system that integrates planning, forecasting and quality control into a coherent, practical model.

There has been a tendency historically to focus the interventions of salt production on specific improvements, such as those of mechanisation or filtration, without considering adequately environmental or operational data, as well as market data. The suggested Integrated Climate-Intelligent Salt Production Ecosystem is thus a breakthrough since it uses an integrated digital platform with advanced environmental forecasting algorithms, adaptive planning models, real-time quality control overseen by computer vision, and federated learning empowered waste valorisation.

The Crystallization Forecasting component, in this ecosystem, is fundamental. It brings the predictive analytics power with critical needs to address the unpredictable threats of weather and inject predictability back into the yield of salt production. Using a state-of-the-art machine learning model and incorporating historical weather data, local environmental conditions, and/or pond-specific measurements, this component enables the operator of salterns to make precise, effective actionable forecasts of the best crystallization and harvesting times. Site-specific, real-time insights allow producers to foresee disruptive climate events, schedule operations based on strategic insight and minimize the risk of failed cycles or waste of resources.

1.1. BACKGROUND AND LITERATURE SURVEY

One of the most ancient industrial processes is the extraction of salt using the method of solar evaporation, which is traditionally observed in the tropical coastal settings, since high temperatures and scarcity of rainfall establish an environment in which salt can easily crystallize [2]. Sri Lanka has major salterns in Hambantota, Puttalam, and Elephant Pass that plays a significant role in supplying salt in the country [3]. However, this has changed drastically over the past few years because of seasonal anomalies and heightened climate variability, which has complicated the conventional production cycles making them adapt to the operations without the use of effective predictive and planning tools [4].

Present operational planning in these salterns relies mostly on manual estimation, and past averages usually become inadequate in unpredictable weather conditions. The absence of integrated decision-support systems makes operators provide crucial decisions based on incomplete information (weather predictions of publicly funded weather agencies, market prices provided by traders, and informal estimates of yield using subjective judgment of experience), leading to inefficiencies and lost reliability in production [5].

The use of artificial intelligence (AI) in production planning has proved to be successful in other countries that are in close reliefs in terms of resource quest. As such, in agriculture, AI can be used to plan crops adaptively based on weather, soil, and markets data to maximise yields [6]. Predictive analytics are used in the fisheries industry, in which it runs on maximizing fish quotas to operate in an unstable environment [7]. Likewise, in manufacturing, the reinforcement learning algorithm is utilized in improving resource scheduling and operation prowess [8]. Such global progress notwithstanding, these AI-driven methodologies have not effectively been modified to deal with the unique interaction of the environmental, economic, and regulatory complexities that exist in the salt industry in developing nations such as Sri Lanka.

Sri Lanka has a salt industry facing two kinds of pressure, which are imposed on it by the market dynamics and by the regulatory frameworks. During domestic trading, the spheres need stable plans and cheap salt, and in communities abroad, traders must meet strict purity standards by international agencies [9]. A respectable operational planning system has to wade through these perhaps competing objectives without compromising on economy, as well as respecting environmental sustainability requirements [10].

In this literature, there is an urgency to use integrated data-driven decision-support solutions involving climatic forecasts and market trends, as well as regulatory requirements that will facilitate the more proactive and efficient operational planning in Sri Lankan salterns, and most importantly in Puttalam. This growth of such systems

holds out the promise of enhancing adaptation to climate variability, maximization of resource utilization as well as quality of products and hence will tackle problems around the globe and locally affecting the industry.

1.2. RESEARCH GAP

It has been seen that even though many countries in the world have developed and implemented AI-driven operational planning within an industry like agriculture, fisheries, and manufacturing, an industry-specific and integrated system with regards to salt production is not found in Sri Lanka. Current digital applications in adjacent industries are concerned either with simple weather notification or narrow areas of operation, rather than building a more complete, multi-parameter decision-making regime, specific to salt production.

The key research gaps identified and addressed by the components of the proposed Integrated Climate-Intelligent Salt Production Ecosystem are:

- **The absence of Multi-Parameter Integration:**
Current approaches to planning in the salt industry do not integrate its main inputs such as climate predictions, changing production capabilities, demand levels, operational expenses, and legal requirements. This shortcoming is filled by Component 1: Crystallization Forecasting, which runs on multimodal data crystallization forecasting that incorporates weather forecasts and other measurements and takes the form of highly accurate data-driven forecasts of optimal crystallization conditions and crop harvesting time.
- **Reactive, as opposed to Predictive Decision-Making:**
Existing methods only make decisions based on the observed environmental changes and not on the ability to predict the changes, which results in a loss of efficiency. This gap is filled by component 2, the Adaptive Intelligent Planning Engine (IPE) which combines forecasts, market trends, cost structures, and regulatory requirements into frame-based predictive operational plans that make proactive management proactive.
- **Failure to Adapt in Real Time to New Conditions:**
The market conditions and salt production environments are very unstable and quick to change with regard to weather and demand. Nonetheless, existing systems do not have the capability of automatically varying plans on an almost real-time basis. This obstacle is circumvented by Component 3, which is Computer Vision Quality Control that facilitates real-time automated monitoring of salt quality feeding real-time feedback into operational changes, and Component 2, continual refreshes of planning strategies using real-time input data.

- Lack of planning tools relevant to the stakeholders:

The business strategy of saltern operators differs; the owners of one saltern are oriented to the production of high-quality salt that covers export needs whereas others want to increase their output volume to sell it into local markets. Nowadays, there are no tools with such possibility. The user input and customization layer (Component 2) builds into the Adaptive Intelligent Planning Engine a customized level of operation strategies in line with the specific priorities of individual stakeholders.

- AI Techniques That Are Not Utilized to the Full Potential in Salt Production Intrigues:

Though AI technologies like reinforcement learning, fuzzy logic, and heuristic optimization are well-developed in other domains, they are not applied to the conditions and the possibilities of salt production in the tropics. The system, where all components are integrated (namely, Component 4: Federated Learning Waste Valorisation), utilizes state-of-practice AI solutions, including federated learning algorithms, to make it possible to coordinate the optimization processes across numerous locations, all the time protecting the privacy of data.

Comparative Analysis of Existing System vs. Proposed Solution

System	Features	Limitations	Our Solution: Improvements
Salt Bae	AI-based salt yield predictions, mobile UI	No regulatory/market integration, lacks privacy frameworks	Compliance, local privacy, regulatory integration
IBM Watson Agriculture	Crop planning, weather analytics, global	Generic; not adapted for salt or tropical microclimates	Salt-specific, local climate and regulation aware
Fisheries AI (Catch Quotas)	Predictive analytics for harvest limits	Marine focus; not for stationary evaporative ponds	Designed for pond production and land processes
Local Manual Practices	Operator experience, logbooks	Lacks data, non-predictive, high risk with climate change	Real-time, predictive, not reliant on experience

System	Features	Limitations	Our Solution: Improvements
Palm Oil FL Platforms	Federated learning for privacy, edge AI	Not applied to salt, lacks market/regs integration	Privacy, plus market/regulation, salt focus

Figure 1.2.1: Comparative Analysis of Existing System vs. Proposed Solution

Coordinating these interconnected gaps through a modular yet integrated ecosystem, this project aims at changing how the Sri Lankan salt industry, specifically the Puttalam salterns, is currently operated, due to the current fragmented and reactive management it is based upon, towards sustainable, data-driven, and resilient operational planning.

1.3. RESEARCH PROBLEM

The key research question that will prove to be the impetus of this venture can be put in the following words:

“How would an adaptive, learnt planning algorithm be formulated to incorporate real time environmental, production, economic factors, and regulatory factors, in the context of generating optimal operational strategies to enhance profitability, efficiencies, and make the salt production industry of Sri Lanka import-dependent with regards to climatic variability?”

This problem statement captures both the technical challenge - creating and deploying an effective, multi-parameter optimization tool that can address complex and dynamic inputs and the practical need, giving salt producers access to sophisticated decision-making tools to respond to an increasingly erratic climate and market environment.

The issue can even be further broken down into three critical dimensions:

- **Environmental Uncertainty:** Dealing with unpredictable changes in weather and climate conditions that affect the salt crystallization process and harvest directly and are largely dependent on changes in temperature, rainfall, and other weather-related conditions.
- **Economic Variability:** Adjusting the production schedule and resource allocation to varying and changing market demand, availability, and price signals in a way that can maximize revenue and minimize wastage and overproduction.
- **Regulatory Compliance:** This is necessary to ensure that operational plans and final products are acceptable as per the strict national and international quality standards and environmental regulations, which are vital in sustaining export market capability and industry sustainability practices.

Solving this complex issue will not only increase the level of self-reliance of salt production in Sri Lanka but also help in increasing the competitive ability of the industry in the local and international markets. Moreover, the plan of developing the solution as a scalable and modular framework will facilitate transferability of the project to other tropical coastal areas of the world, which might be similar to those of the project areas in terms of environmental and operational similarities in salt production.

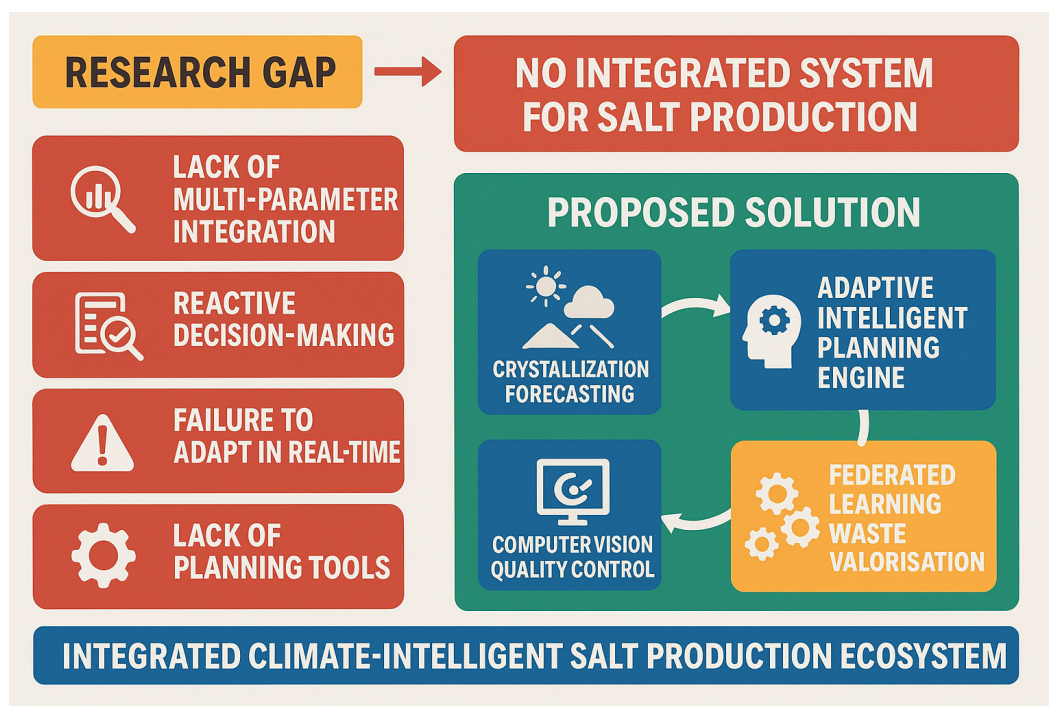


Figure 1.3.1: Research Gap

2. RESEARCH OBJECTIVES

This section introduces the overall aims of the Integrated Climate-Intelligent Salt Production Ecosystem as a multi-component platform to modernize, optimize and future-proof climate-sensitive coastal salt production in Sri Lanka.

2.1. MAIN OBJECTIVE

In producing and deploying a bespoke, AI-powered ecosystem designed to support the salt production industry in Sri Lanka by facilitating predictive environment analytics, adaptive operating program code, autonomous quality control, and joint optimization of operations resources, which works toward increased productivity, sustainability, and climate adaptation, along the entire salt production value chain.

SMART Breakdown:

- **Specific:** Design and implement an integrated platform that would consist of four autonomous but interconnected AI-powered modules (crystallization forecasting, intelligent planning, computer vision quality control, federated learning waste valorization).
- **Measurable:** The performance of the success is measured through increases in production yield ($\geq X\%$), use efficiency in resources ($\geq Y\%$), quality product compliance ($\geq Z\%$), and the decrease of wastes as well as positive feedback on at least two operational pilots.
- **Achievable:** Achievable by leveraging the project team's expertise in AI, software engineering, and salt industry operations, and by collaborating with local saltern partners for piloting and validation.
- **Realistic:** Realistic to the available technology and ability to access experienced personnel in operation, besides high stakeholder involvement in the industry.
- **Time-bound:** To get fully implemented, deployed, and reviewed even on a project level.

2.2. SPECIFIC OBJECTIVES

The specific objectives below specify the nature and extent of the contributions of each ecosystem component as well as promoting synergy among the components:

- **Crystallization Forecasting:** To develop and validate machine learning models of weather, sensor, and historical production data, and give early and precise deep forecasts of optimal crystallization and harvest windows.
- **Intelligent Planning Engine:** Develop flexible AI-based planning systems that can take predictive forecasts, real-time market, manufacturing, and regulatory information, and convert it into stakeholder-optimized, actionable operational plans to make profitable, proactive decisions.

- **Computer Vision Quality Control:** Implement a high-throughput automated computer vision system to grade the salt and detect salt impurities in real-time so that the high-quality and regulatory standards of the product can be achieved, and any human touch and production bottlenecks are reduced.
- **Federated Learning Waste Valorization:** Develop a federated learning infrastructure, a secure and collaborative setting to optimize the reuse and valorization of salt waste in a distributed set of sites, without compromising data that is sensitive but also enhancing a sustainable circular and economic enrichment initiative.

In attaining these goals, the project will not only assist Sri Lanka in the self-sufficiency in salt production, competitiveness in both regional and international markets, and sustainable and technology-driven management of its resources, but also helping provide the country with the necessary leadership in the above aspects. The replicability of the ecosystem in terms of modularity and scalability also allows the ecosystem to be implemented in other tropical coastal nations that are grappling with the same challenges and thereby making more contributions to the topic of climate-smart industry transformation.

3. METHODOLOGY

3.1. System Overview

The Integrated Climate-Intelligent Salt Production Ecosystem is a modular, AI-driven platform (potentially modular) that aims to revolutionize the salt production industry in Sri Lanka by performing end-to-end decision support tasks to the entire salt production lifecycle. The structure of its architecture is divided into four related components that take care of each of the vital steps in operational planning, execution, and optimization.

System Components

1. **Crystallization Forecasting:** Combines multimodal information (weather and satellite imagery in addition to historical production information) with advanced time series models (ARIMA, Prophet, and LSTM) to determine the ideal conditions to crystallize salt and harvest it. This makes it possible to develop proactive plans and to reduce the risks of yields depending on the uncertainty of the climate.
2. **Intelligent Planning Algorithm:** Converts predictive intelligence of Component 1, with real-time market trends, production capacities, economic data, and regulatory requirements into optimized operating strategies implemented into actions. It uses multi-parameter optimization, heuristics, fuzzy logic, decision trees, and reinforcement learning to support dynamic data-driven planning, which can change as circumstances change.
3. **Quality Control: Computer Vision:** Automatically grades the salt quality using computer vision image analysis and impurities are detected in real time. This improves management without the need of manual check-up, thereby making it more regulated and assuring the quality of the products to both the local and international markets.
4. **Federated Learning Waste Valorization:** Facilitates the distributed and privacy-preserving coordination of different sites of the saltern to optimize by-product reuse and valorization. By using federated learning and differential privacy, they can be sure that critical operational data will not be shared directly, but it will lead to innovation and data safety.

Key Features

- Modularity
- Real-time Data
- User-Centric Dashboard

Such an interconnected ecosystem has the unique opportunity to consider the complex climate sensitive operation realities of the salt industry in Sri Lanka.

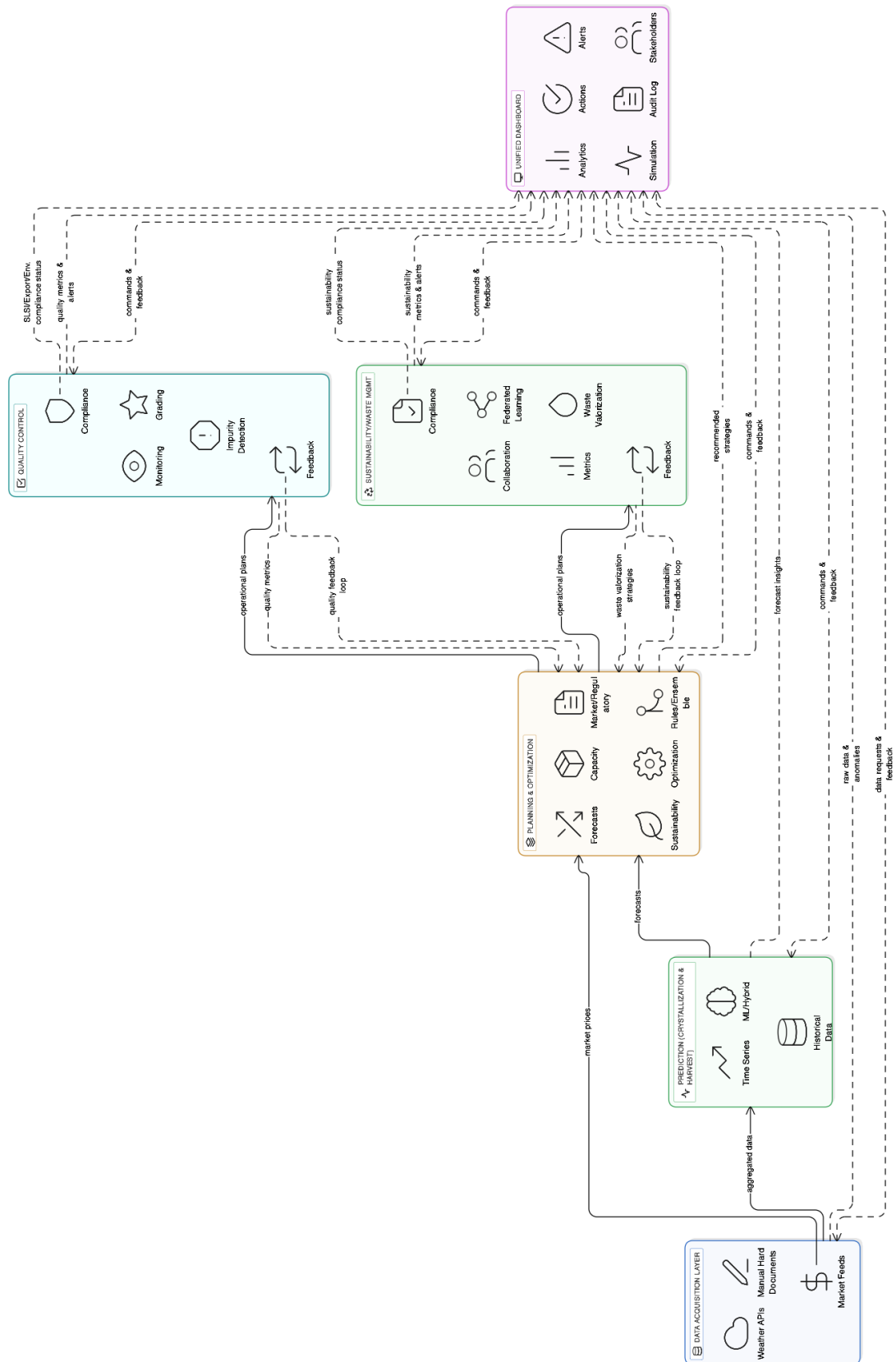


Figure 3.1.1: System Architecture Diagram

3.2. Individual Component

The Crystallization Forecasting system is thought of as a disruptive data-driven predictive analytics tool solving the very weakness of the Sri Lankan salterns' unpredictable and climate-related yield cycles. What happens in practice, operators are compelled to use their past experience or averages and use intuition when they encounter unpredictable rain, temperature, and wind. Past endeavors of production planning in the area are manual in nature and lack connection with environmental science and advanced forecasting [5], AI applications such as Salt Bae in other parts of the world have shown that significant improvements in efficiency and efficacy can be achieved when there is a merger between multi-modality data and machine learning with daily planning. This forecasting framework fills this gap in knowledge because the salt industry of Sri Lanka currently lacks a functional decision-support tool that integrates in a methodical manner production data, local meteorological observations, and remote sensing. This framework provides the legitimacy of model selection on the uniqueness of real Sri Lankan datasets alongside the critical evaluation of the consideration of agricultural and climate forecasting around the world [11] [6]

3.2.1. Requirement Gathering and System Design

To create a powerful forecasting platform without the use of an IoT infrastructure at the first stage, the system will use a wide range of available sources of data:

- History of Production Table Salt production levels, gathering times/periods, and rain-associated losses are extracted as digital data by the physical logbooks kept by saltern managers. The records constitute the backbone of the knowledge about how environmental conditions can be used in the production over the years and seasons.
- Weather Data: Rich datasets on real-time and historical meteorological conditions-such as temperature, rainfall, humidity, and wind speed - will be obtained at the Sri Lanka Department of Meteorology as well as through reliable international providers (NOAA, Metastatic). The records are vital to enable simulations of the environmental forces behind the crystallization and evaporation processes.
- Supplementary Data: Miscellaneous variables such as well-known climate cycles (monsoons), peculiar seasonal variations, and demand proxies will also be collected to facilitate a contextual interpretation and scenario analysis-making sure the forecasting model is locally pertinent and operationally aligned.

Collected and together, these datasets are subjected to an intensive digitization, annotation, validation, and standardization process and set a high-quality baseline to subsequent model development.

3.2.2. Data Processing and Feature Engineering.

Preprocessing effectiveness and aimed at creating the predictive features, is crucial to the success of the forecasting engine:

- **Cleaning:** Information records and weather records may have been incomplete, or have organizationally anomalous entries or consistencies, that must be recognized and identified systematically. This will help eliminate the issue of news that is dirty and untrustworthy as far as the forecasting process is concerned.
- **Normalization:** Parameters of weather data that originated in various servers and geographical positions are normalized to allow comparison with data gathered in unusable periods and at disparate points of data access, to allow a correct model to be Michael Cirino to determine the desired range and impact of each variable.
- **Feature Extraction:** The drivers of crystallization are more directly caught in the derived variables:
 - **Evaporation Index** is a combination of temperature, humidity, and wind speed to determine the strength of evaporation.
 - **Salinity Potential:** Takes rainfall and land dryness into considerations, deductions about whether ponds are near-optimal salinity levels to crystallize.
 - **Lagged Variables:** Moving averages and trends (e.g. over 7 or 30 days) to show the beginning of a favourable or risky period.
- **Data Fusion:** Data collected on weather, satellite, and operational are fused into one common dataset-the richness is maximized and the model training becomes fully data-driven

3.2.3. Forecasting Model Development

The use of a multi-model approach promotes the robustness of the technology and the applicability in context:

- **Statistical Models:**
 - **RIMA:** An Ordinary benchmarking model is useful for stationary trends, especially on bounded trends.
 - **Prophet:** Provides excellent model discovery capability to detect underlying seasonally and trend cycles, as is the case with salt cycles which are monsoon-driven.
- **Artificial Intelligence Models**
 - **Random Forest Regressors:** Process non-linearity and multi-collinearity that are usually prevalent in environmental data.
 - **XGBoost:** Chosen due to its Calibration capabilities across Tabular and Hybrid data that often prove to be best compared to the use of single algorithms in real-life competitions.

- Radio Station Neural Networks
 - LSTM/GRU: The best current option to input complex, sequential patterns particularly in climate records that are long.
- Ensembles: Final predictions are calculated through the combination of individual predictions that have been verified in literature [8], to create more reliability and reduce exposure to individual model bias.

Technological decisions are fully justified by the satisfactory results they display in climate/time-series predictions in terms of the accuracy of the project on hybrid compilation of climate/time-series forecasting.

3.2.4. Model Training, Tuning, and Evaluation

Training, Validation: Training in the models is performed on a window of historical production, environmental data, with cross-validation and withheld out-of-periods when rigorous assessment is achieved.

Metrics Used:

- RMSE/MAE: Measure accuracy and mean error.
- R² score: Display the amount of variance explained by each model.

Hyperparameter Tuning: Implements grid search and Bayesian optimization, thus each model is performing at optimal capacity, and scheduled retraining is based on the accumulation of data.

Technology Assessment: The continuous benchmarking will take place not only against the routine manual/empirical practices of the past, but also against other algorithms that are available within the competing algorithmic approaches and will select not only in regards of accuracy but also in terms of interpretability and within the ability to implement within Sri Lankan saltern operations.

Self-analysis will occur in feedback by operators at pilot locations and success will be gauged as quantitatively as well as by follow-up end-user adoption and confidence.

3.2.5. Dashboard and Visualization Layer

User-Centered Design: Forecast reports can be presented on an easy-to-read dashboard that contains: Daily/weekly risk levels

Easy to understand graphics (e.g. traffic light display, trend lines), and Historical accuracy and confidence forecast overlays.

Feedback Capture: Operators are also able to comment on outputs, mark odd situations, and propose changes to model retraining, so that the dashboard continues to improve with the needs of users and operational realities.

Operational Value: The system streamlines a process that enables the quick translation of complicated environmental analytics into usable immediately accessible information that informs the decision-making process of saltern managers.

3.2.6. Testing and Validation

- Back testing:

The historical simulation of a period beyond two years of data offers rigor in terms of determining whether the system can minimize crystallization loss and reduce outliers weather.

- Pilot Trials:

Lagged live deployments in Puttalam saltern partners not only give a measure of the accuracy of the forecasts but also of their effect on operation, and on user acceptance.

- Stakeholder Review:

Integral participation of stakeholders in the dashboard build and regular survey maintains relevance on the future requirement and targeted approach to the long run adoption.

To sum up, the Crystallization Forecasting system is a well-thought-out attempt to close a real, justified knowledge and operational gap, based on an appropriately selected means of modern data science and artificial intelligence, with the extensive use of both statistic and field-based self-evaluation plans. The given approach guarantees its originality, feasibility, and good business prospects of commercialization in the Sri Lankan salt market and other places.

3.2.7. Work Breakdown Structure

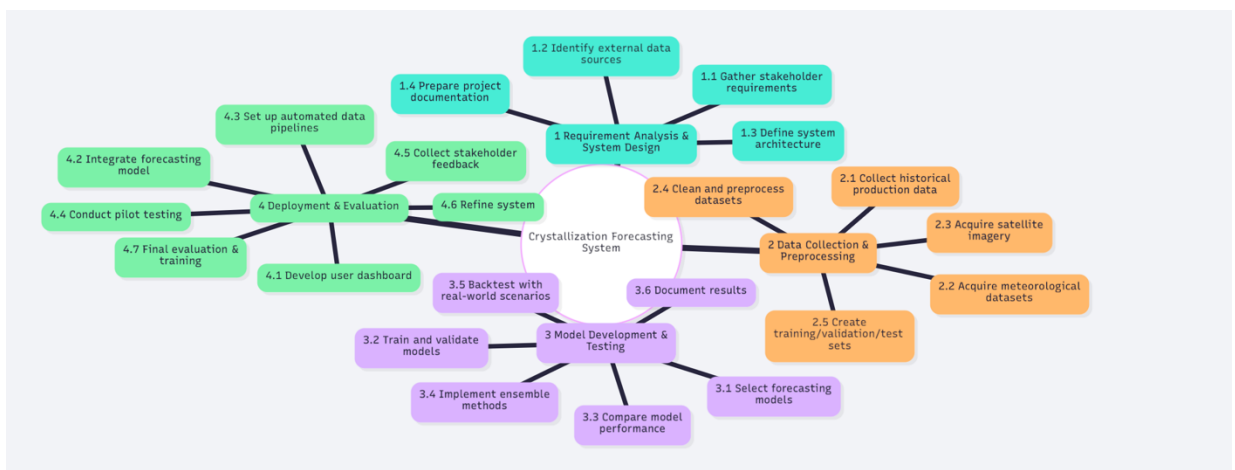


Figure 3.2.7.1: Diagram to represent Work Breakdown Structure

3.3. Materials Required to Carry Out the Project

The functional outcome of the Crystallization Forecasting system design and implementation will be achieved with an array of hardware, software, and strategic resources that fit the requirements of data-oriented multi-source modeling in the conventional salt industry environment.

3.3.1. Equipment

Development Workstations:

One or more laptop or desktop machine per team member, each with at least 16GB RAM. Though fast to develop/train/benchmark at the initial stages, we recommend access to at least one GPU-enabled machine (or cloud GPU resources) during the development, training, and benchmarking of deep learning models (LSTM/GRU).

External Storage Devices

Large-capacity (≥ 1 TB) external disc drives or encrypted solid-state drives to safely archive scanned handwritten logbooks, tens of thousands of historical weather, and any intermediate-processed data.

Reading Aids:

Ability to access a high-resolution camera or scanner to effectively digitize physical production records, weather diaries, and hand-labeled operator logs, able to support an effective data entry and preprocessing pipeline.

3.3.2. Software and Computer Support

- Programming Languages:
 - Python: The first Programming and Modeling environments: The programming language to be used as the core is Python
 - Data Science Libraries: scikit-learn, statsmodels to cover the statistical/modelling basics; TensorFlow and PyTorch to cover advanced deep learning pipelines.
 - Data Management: pandas, and NumPy to clean, wrangle and engineer features.
- simulations and viewing:
 - Plotly, Matplotlib and Seaborn to create insightful, interactive plots, heat maps, and comparative analytics used in model evaluation and operator facing dashboards.
- Database Systems:

At the data-structure- and cleaning-only layer, a PostgreSQL database was installed to maintain the structured, cleaned datasets-logbook entries, daily weather data, and model outputs-with fast and reliable qualities of data-retrieval and integrity.

- Dashboard design and development

React.js will be used to develop an interactive and responsive dashboard where saltern operators and stakeholders will be able to afford access to daily crystallization forecasts, historic analytics and risk alerts. The two frameworks are strong in terms of science, and user-centric applications.

- Collaboration, Version Control
 - GitHub with version control, code collaboration and issue tracking, a means of ensuring the smooth development of distributed team members.
 - Trello or any other agile project management tool to assign and distribute workload, visualize progress and track milestones.

3.3.3. Other Resources

- External Access to Data
 - Weather APIs: Availability of historical and real-time APIs like OpenWeatherMap and Meteostat to perform multi-provider data aggregation of meteorology data to support redundancy and cross-validation.
 - Reference Datasets: Ideally, secondary datasets of the Sri Lanka Department of Meteorology, climate archives of local universities or published benchmark sets to the monsoon and drought modelling.
- Translation and Data Entry Support:

Secure access to Sinhala/Tamil translators and trained data-entry staff to rapidly, and accurately, convert handwritten records into a digital format--reducing data-loss and guaranteeing attention to site-specific details.

- Stakeholder Engagement:

Permanent assistance of the nearby saltern managers, operators, and the meteorological department in clarification, feedback, and data validation eliminates context inappropriateness and makes user-centered design even easier.

- Security and Backup:

Store raw and processed datasets on secure cloud storage (e.g. Google Drive) which is regularly backed up to limit damage, loss of data due to hardware failure, etc.

3.4. Data Required for the Project and Collection Methods

The achievement of the Crystallization Forecasting system lies in the strength and quality of the dataset. The following data type, sources, and methods of its collection have been most diligently chosen to monitor a full scale of environmental and operationally related variables that affect the outcome of the salt crystallization process in Sri Lankan salterns.

- Historical Salt Production - Local saltern logbooks (in Sinhala/Tamil)
 - Digitization & Manual Entry: Physical handwritten records are scanned and manually transcribed into digital format for analysis. Ensures detailed temporal and spatial coverage of past yields, harvesting dates, and loss events.
- Weather Data - Sri Lanka Department of Meteorology, NOAA, Meteostat
 - Integration With Current Weather History API: Official agencies supply weather data (pinpointing time) (temperature, rain, humidity, wind) via APIs or downloadable files. Historical weather is very important to modeling and verification.
- Climate Cycles & Events - Published scientific and government climate reports
 - Digitization & Manual Entry: Physical handwritten records are scanned and manually transcribed into digital format for analysis. Ensures detailed temporal and spatial coverage of past yields, harvesting dates, and loss events.
- Supplementary Operational Data - Saltern staff, supervisors
- Data Digitization and Quality Assurance:
 - Translation: Sinhala/Tamil documents will also be translated and communized into model inputs.
 - Cross-Verification: Secondary sources of data and interviews will be used to cross-check with production, weather and operator records to reduce error.
 - Standardization- All data should be in coherent formats (e.g., temperature in Celsius, rainfall in mm, yields in kg/tons) in order to be appropriate in modeling.

Justification:

Such a multi-pronged design will provide coverage of all critical variables, it exploits the expertise and local knowledge coupled with scientific measurement, and offers high data integrity. This is necessary to ensure solid and reliable forecasting models are created in the scenario of a combination of structured data and unstructured legacy sources.

3.5. Estimated Time Frame

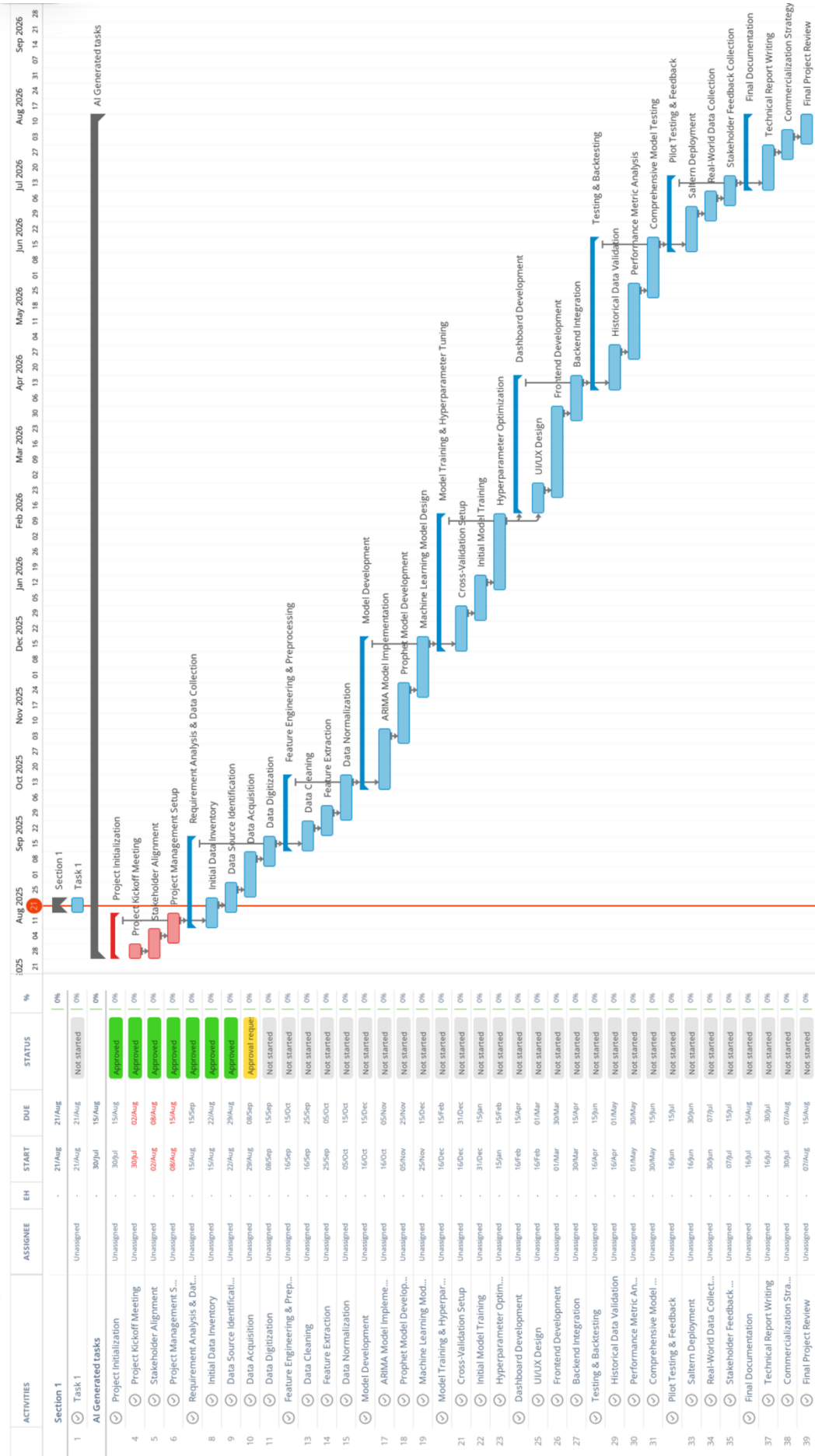


Figure 3.5. 1: Gantt Chart for Crystallization Forecasting Component

3.6. Commercialization

The Crystallization Forecasting component, with its origins in academic research excellence, has a good commercialization prospect in the salt industry in Sri Lanka and in coastal tropical countries around the world. Accurate, reliable predictability of yields despite climate variability creates operational resiliency for saltern operators and economic value by providing policymakers and investors with confidence in project planning, financing, and return on funds invested.

3.6.1. Product Differentiation and Value Proposition

Differentiation:

- **Local Creation Weather Forecasting:** Localized weather forecasting, unlike generic or universal weather forecasting, is specifically calibrated to the saltern environment of Sri Lanka and provides hyper-specific and highly predictive crystallization-readiness Forecasting.
- **Combination of Multimodal Data:** It is the only model that combines historical production data, weather, and environmental data into a single AI-based predictive model, improving the accuracy of forecasting as compared to heuristics.
- **AI-Powered Precision:** The component uses machine learning and deep learning algorithms, which are proven to be superior to human-based decisions made on experience and are dramatically lowering business operation uncertainty.
- **Actionable Insights:** Data on complex environmental conditions is distilled into simple-to-understand and easy-to-interpret harvesting indicators, such as a traffic-light system of alerts (Red/Yellow/Green) to enable informed and timely operator action.

Value Proposition:

- **To Salt Producers:** Reductions in losses because of scattered rain events, better harvest planning, higher quality of yield quality, and higher profitability.
- **To the Policy Makers and Regulators:** More robust modelling and forecasting tools allow enhanced planning of national salt supply and contribute to compliance with export qualities requirements.
- **To Investors and Stakeholders:** A scaled and modular system designed to have possible use in diverse tropical and sub-tropical coastal locations maximizes overall investment potential and serves to address potential climate risks.

3.6.2. Revenue Model

- **Subscription-Based (SaaS):**
 - Licensed operators of saltern who subscribe with the seasonal or monthly services do so to get forecast services.

- The makings of tiered offerings such as Basic (standard weather + 7-day forecasts) and Premium (extended seasonal outlooks, risk alert, and advanced scenario testing) are provided.
- Data-as-a-Service (DaaS):
 - Supply chain participants- exporters and suppliers of goods- buy collective and anonymized forecasts of production to maximize transport and stabilization of prices.
- Forging Government and NGO Partnerships:
 - Partnerships with the Ministry of Fisheries and development agencies can allow deploying on a wide scale, funded at the level of the climate adaptation and livelihoods support program.
- Consulting Services:
 - Large-scale industrial saltern customized system deployment, training, and integration packages to operators and commercial clients.

3.6.3. Scalability and Market Penetration

Phase 1: National Scale: Initially, 2 salterns in Sri Lanka will be targeted, namely Hambantota and Puttalam, to have maximum impact and demonstration potential.

Phase 2 Regional Expansion: Adaptation of salt industries of neighboring countries of South Asia (India, Bangladesh) with similar climatic issues.

Global Reach Phase 3: Expand into tropical salt-producing regions of Southeast Asia, Africa, and Latin America, where demand is growing in salt production solutions that are adaptable of climate change.

Scalability Enablers:

- Cloud-native design enables the flexibility and low-latency deployment and use across multiple sites.
- API-driven and modular design allows expansion to encompass future IoT sensor information and more modules of the ecosystem.
- Continuous model retraining and enrichment of data increases the accuracy of forecasts as user bases increase.

3.6.4. Marketing Strategy

Pilot Demonstrations: Place and test a forecasting system with government-managed salterns to create a mission case study.

Stakeholder Engagement: Seek partnerships between the National Salt Corporation of Sri Lanka and local producer cooperatives to gain the stakeholders' trust and increase adoption.

Education and Training: Carry out seminars and thorough training on the use and interpretation of the system to those that are operating salterns and managing them.

Online Marketing: Use the industry related groups, online technology in agriculture and professional sites such as LinkedIn to reach the investors and third parties.

Government Policy Alignment: Develop a message that positions forecasting system as an innovative technique in adapting to the changing climate, and solicit the support of line ministries and agencies.

3.6.5. Intellectual Property Protection

Copyright in software Registration of ownership of algorithm code and user interfaces.

One - Data Protection: Confidentiality of historical and compiled data on salt production and environmental records are used as trade secrets.

Patent Potential: A survey of the potential to patent protection around the new integration approaches based on multimodal environmental data provided to predictive crystallization forecasts.

4. PROJECT REQUIREMENTS

The effectiveness of the Crystallization Forecasting component will rely on an appropriately shaped set of requirements related to the technical capacity of the system, user requirements as well and operational limitations. These fall into the categories of functional, user, system, and non-functional requirements with specific use cases and test cases to direct design and verification.

4.1. Functional Requirements

- Data Management

R1: Archive, digitize, and store old records on salt production in terms of a classified and searchable database.

R2: Add weather data (temperature, rainfall, humidity, wind) to trusted weather APIs.

R3: Process, pre-process, and integrate satellite-based environmental measures, like evaporation rates and land dryness.

- Forecasting

R4: Produce precise short periods of crystallization readiness of multiple saltern ponds.

R5: Present seasonal forecasts (13 months): Distribute seasonal schemes with a pattern considered for long-term variations in climate.

R 6: Accompany predictions with levels of confidence or probabilistic estimates of the reliability of predictions.

- Visualization & Dashboard

R7: See forecasts through an easy-to-navigate dashboard that shows trend graphs, readiness, and risk alerts. Use an understandable system of the readiness of harvest (e.g., a traffic-light system (Red, Yellow, Green)).

R9: Allow export/ download forecast reports in standard formats (PDF, CSV).

- Validation & Feedback

R10: Conduct a backtests of the forecast compared with the historical production data to assess model performance.

R11: Provide the operator dashboard with measures of the accuracy of the presented forecasts (e.g., RMSE, MAE).

R12: Allow end users to submit feedback about the quality of the forecasts to be able to develop the forecasts over time.

4.2. User Requirements

- **Saltern Operator Requirements (Primary Users)**
 - Our convenient, non-technical interface to daily forecasts of crystal formation will be an immediate benefit.
 - There are clear indicators alerting the best time to go out to harvest.
 - Access to forecasts, views of the past, and the future season to aid long-term planning.
 - The lengths of training required are short with easy-to-understand results.
- **Industry Stakeholder Requirements**
 - Aviation forecasts at the aggregated/regional level are deemed relevant to policy and regulation.
 - Export capability of creating forecast reports to assist the downstream supply chain and pricing decisions.
 - It helps in gaining insights on the levels of future salt yield fluctuations so as to predict market trends.

4.3. Non-Functional Requirements

Scalability: The system should be able to easily cope with growing amounts of digitized records and increased sales coverage

Accuracy: The forecast accuracy should exceed the baseline heuristic, and the goal should be over 75 percent correctness as measured.

Reliability: High availability of the system performance is required, with a system uptime of 99 percent.

Usability: The interfaces should allow ease of use and very little technical expertise.

Security: Securing API calls (HTTPS), authenticating users, and encrypting the data in storage are required.

4.4. Test Cases

Table 4.4.1: Test Cases

Test Case ID	Description	Input	Expected Output	Pass Criteria
TC01	Import weather API data	API key, request params	Valid JSON response with weather data	200 response, correct data returned
TC02	Upload historical production data	Historical production data file	Data normalized and ready for modeling	Normalized data available for modeling
TC03	Generate 7-day forecast	Weather and production data	Forecast displayed with confidence levels	Forecast with confidence levels displayed
TC04	Generate seasonal outlook	Historical weather data	Long-term trend graph shown	Trend graph correctly rendered
TC05	Trigger rainfall alert	Rainfall threshold data	Dashboard displays “High Risk – Rain Expected”	Alert message displayed on dashboard
TC06	back test with historical data	Historical data set	RMSE and MAE metrics displayed	Metrics (RMSE, MAE) correctly calculated
TC07	Operator feedback submission	Feedback input form	Feedback stored and linked to forecast record	Feedback stored with correct linkage

4.5. Wireframes

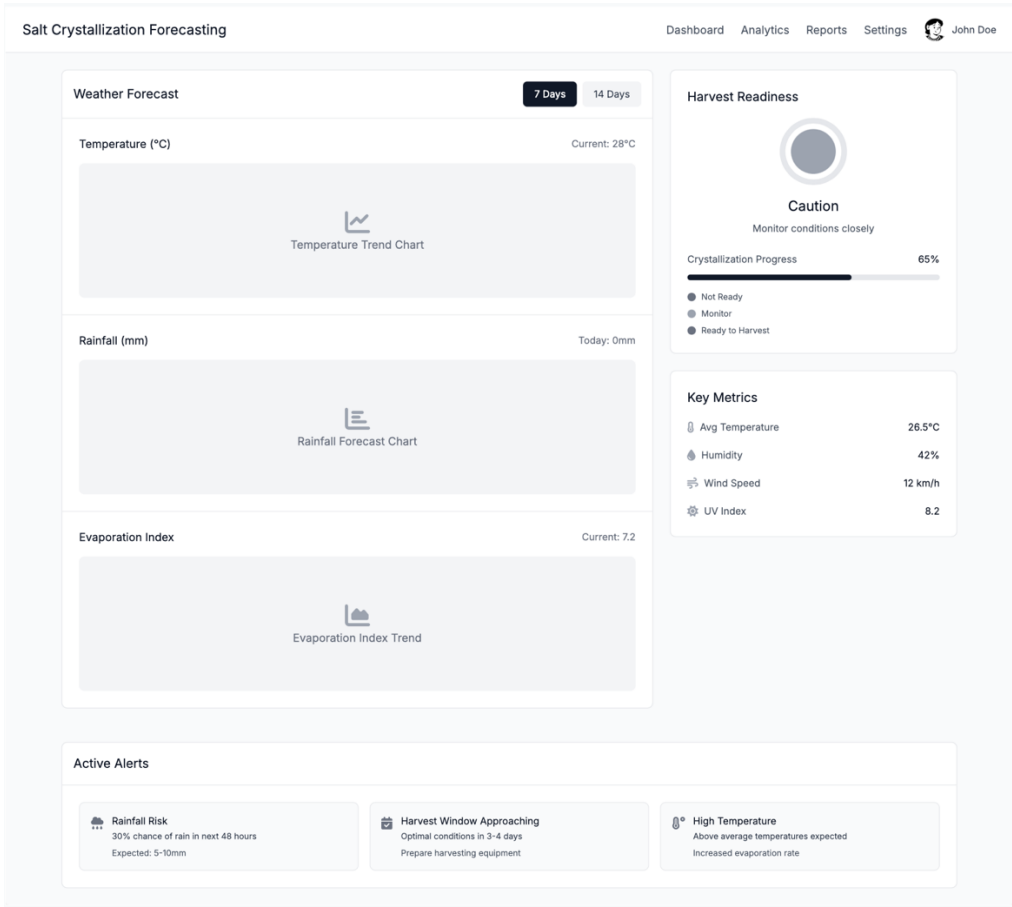


Figure 4.5.1: Wireframe for salt crystallization dashboard

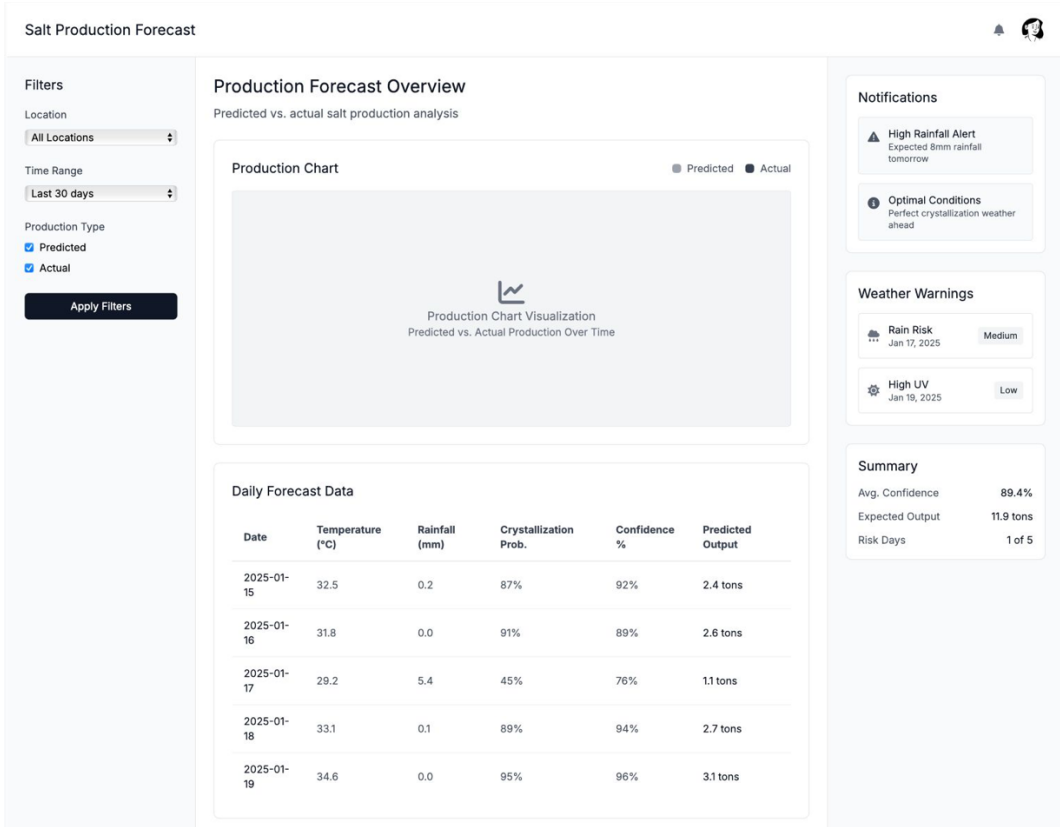


Figure 4.5.2: Wireframe for Salt Production Dashboard

5. BUDGET AND BUDGET JUSTIFICATION

Expense	Cost (LKR)	Net Total	Justification
Personal Costs			
Travelling	36,000	36,000	3000 x 12 covering site visits to Puttalam and supervisor meetings.
Software And Service			
Domain Purchase	1,000		4000 / 4 mem
Host and deploy	17,500		AWS EC2 Instance (200\$) 70,000/4
Perplexity Pro	18,000		6000 x 12 / 4
Premium Api	12,500		For data gathering (e.g.: weather) 50,000 / 4
Google Collab Pro	36,000		For model training (10\$ = 3000 per month)
Cloud Storage	20,000	105,000	Additional storage if needed
Grand Total		141,000	

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7. APPENDICES